Lecture exercise branching

This notebook was generated in student mode.

Exercise: Positive, Negative, or Zero

Positive, Negative, or Zero

Write a program that determines whether a given number is positive, negative, or zero.

Use the number `num = -5` and store the result in variable `result` ("positive", "negative", or "zero").

```
num = -5
# Write your solution here
result =
```

```
assert result == "negative"
```

Python Tutor Visualization

The following cells contain code to generate links to Python Tutor for visualizing this exercise. You need to first run the cell containing your solution code before running the cell containing the Python Tutor link.

```
from IPython.display import HTML;import urllib.parse as u;HTML(f'<a href="https://pythontutor
```

Exercise: Even or Odd Number

Even or Odd Number

Write a program that determines whether a given integer is even or odd.

Use the number `num = 7` and store the result in variable `result` ("even" or "odd").

```
num = 7
# Write your solution here
result =
```

```
assert result == "odd"
```

Python Tutor Visualization

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```
from IPython.display import HTML;import urllib.parse as u;HTML(f'<a href="https://pythontutor
```

Exercise: Grade Evaluator

Grade Evaluator

Write a program that converts a numerical score (0-100) to a letter grade:

- 90-100: A
- 80-89: B
- 70-79: C
- 60-69: D
- 0-59: F

Use the score `score = 85` and store the letter grade in variable `result`.

```
score = 85
# Write your solution here
result =
```

```
assert result == "B"
```

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```

Exercise: Divisibility Checker

Divisibility Checker

Write a program that checks if a given number is divisible by both 3 and 5.

Use the number `num = 30` and store the result in variable `result` ("divisible by both" or "not divisible by both").

```
num = 30
# Write your solution here
result =
```

```python
assert result == "divisible by both"
```

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```python
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```

Exercise: Leap Year Checker

Leap Year Checker

Write a program that determines whether a given year is a leap year or not.

A leap year is:

- Divisible by 4 but not by 100, OR
- Divisible by 400

Use the year `year = 2024` and store the result in variable `result` ("leap year" or "not leap year").

```python
year = 2024
# Write your solution here
result =
```

```python
assert result == "leap year"
```

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```python
from IPython.display import HTML;import urllib.parse as u;HTML(f'<a href="https://pythontutor
```

Exercise: Vowel or Consonant

Vowel or Consonant

Write a program that determines whether a given letter is a vowel or consonant.

Use the letter `letter = "A"` and store the result in variable `result` ("vowel" or "consonant").

```python
letter = "A"
# Write your solution here
```

```
result =
```

```
assert result == "vowel"
```

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```
from IPython.display import HTML;import urllib.parse as u;HTML(f'<a href="https://pythontutor
```

Exercise: Password Validator

Password Validator

Write a program that validates a password. If the password is "password123", grant access, otherwise deny access.

Use the password `password = "password123"` and store the result in variable `result` ("Access granted" or "Access denied").

```
password = "password123"
# Write your solution here
```

```
result =
```

```
assert result == "Access granted"
```

Python Tutor Visualization

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```
from IPython.display import HTML;import urllib.parse as u;HTML(f'<a href="https://pythontutor
```

Exercise: Largest of Three Numbers

Largest of Three Numbers

Write a program that finds the largest of three given numbers.

Use the numbers `num1 = 15`, `num2 = 8`, and `num3 = 22`. Store the largest number in variable `result`.

```
num1 = 15
num2 = 8
num3 = 22
```

```python
# Write your solution here
result =
```

```python
assert result == 22
```

Python Tutor Visualization

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```python
from IPython.display import HTML;import urllib.parse as u;HTML(f'<a href="https://pythontutor
```

Exercise: Day of the Week

Day of the Week

Write a program that converts a number (1-7) to the corresponding day of the week:

- 1: Monday
- 2: Tuesday
- 3: Wednesday
- 4: Thursday
- 5: Friday
- 6: Saturday
- 7: Sunday
- Any other number: "Invalid number"

Use the number `day = 5` and store the day name in variable `result`.

```
day = 5
# Write your solution here
result =
```

```
assert result == "Friday"
```

```
# Test with invalid number
test_day = 8
if test_day == 1:
    test_result = "Monday"
elif test_day == 2:
    test_result = "Tuesday"
elif test_day == 3:
    test_result = "Wednesday"
elif test_day == 4:
```

```
    test_result = "Thursday"
elif test_day == 5:
    test_result = "Friday"
elif test_day == 6:
    test_result = "Saturday"
elif test_day == 7:
    test_result = "Sunday"
else:
    test_result = "Invalid number"
assert test_result == "Invalid number"
```

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```

Exercise: Simple Calculator

Simple Calculator

Write a program that performs basic arithmetic operations (+, -, *, /) on two numbers based on the given operation.

Use `num1 = 10.0`, `num2 = 3.0`, and `operation = "+"`. Store the result in variable `result`. Handle division by zero with an error message.

```
num1 = 10.0
num2 = 3.0
operation = "+"
# Write your solution here
result =
```

```
assert result == 13.0
```

Python Tutor Visualization

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```
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```

Exercise: Triangle Type Classifier

Triangle Type Classifier

Write a program that determines the type of triangle based on the lengths of its three sides:

- Equilateral: All sides are equal
- Isosceles: Two sides are equal
- Scalene: No sides are equal

Use `side1 = 5.0`, `side2 = 5.0`, and `side3 = 8.0`. Store the triangle type in variable `result`.

```
side1 = 5.0
side2 = 5.0
side3 = 8.0
# Write your solution here
result =
```

```
assert result == "isosceles"
```

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```
from IPython.display import HTML;import urllib.parse as u;HTML(f'<a href="https://pythontutor
```

Exercise: Maximum of Three Numbers

Maximum of Three Numbers

Write a program that finds the maximum (largest) of three given numbers.

Use the numbers `num1 = 12.5`, `num2 = 8.3`, and `num3 = 15.7`. Store the maximum number in variable `result`.

```
num1 = 12.5
num2 = 8.3
num3 = 15.7
# Write your solution here
result =
```

```
assert result == 15.7
```

```
# Test with different maximum
test_num1, test_num2, test_num3 = 25.0, 18.5, 20.2
if test_num1 >= test_num2 and test_num1 >= test_num3:
    test_result = test_num1
elif test_num2 >= test_num1 and test_num2 >= test_num3:
    test_result = test_num2
else:
    test_result = test_num3
assert test_result == 25.0
```


Python Tutor Visualization